FOREST AND WILDLIFE RESOURCES

Geography - Class 10 | Chapter 2 | Deep Structured Notes

1. Introduction to Forest and Wildlife Resources

Why Forests and Wildlife are Important

Forests and wildlife are essential components of the ecosystem and vital for human survival. They provide oxygen, clean air, water, and habitat for countless species. Forests are the lungs of the Earth and play a crucial role in maintaining ecological balance.

Forests and wildlife provide numerous benefits to humans:

- Ecological benefits - maintain biodiversity, prevent soil erosion, regulate climate, and purify air and water.
- Economic benefits - provide timber, fuel, fodder, medicine, food, and income through tourism.
- Cultural and spiritual benefits - hold religious and cultural significance for many communities.

- Scientific benefits - serve as sources of genetic material for research and development.

Biodiversity refers to the variety of life forms on Earth, including plants, animals, microorganisms, and their ecosystems. India is one of the 12 mega-biodiversity countries in the world, with enormous variety of flora and fauna.

India's Rich Biodiversity

India has about 8% of the world's total biodiversity despite having only 2.4% of the world's land area. This makes India one of the most biodiverse countries in the world.

India has approximately 81,000 species of animals and 47,000 species of plants. This includes:

- About 15,000 flowering plants and many endemic plant species.
- About 90,000 species of animals including mammals, birds, reptiles, amphibians, and fish.
- Thousands of species of insects, fungi, and microorganisms.

Many species are endemic to India, meaning they are found nowhere else in the world. Examples include the Asiatic lion in Gujarat, Indian rhinoceros in India and Nepal, and many unique species of birds, reptiles, insects, and plants.

India's biodiversity is under threat due to habitat loss, poaching, pollution, and climate change. Many species are endangered or extinct due to human activities. Conservation efforts are essential to protect this rich biodiversity.

2. Types of Forests in India

Classification of Forests by the Indian Forest Service

The Indian Forest Service classifies forests into three main categories based on ownership and management.

Reserved Forests are considered most valuable for conservation. These forests are reserved mainly for protection and controlled use. Human activity is restricted and permission is required. More than half of India's forest area is classified as reserved forests.

Protected Forests are protected from further degradation. Limited human activities such as grazing and collection of firewood may be allowed under rules. About one-third of India's forest area is protected forest.

Unclassed Forests include forests and wastelands belonging to the government, communities, or private individuals. These forests often require better management and protection.

Classification Based on Vegetation and Climate

Tropical Evergreen Forests, also called Tropical Rainforests, are found in areas with heavy rainfall above 200 cm and a short dry season. They remain green throughout the year because different trees shed leaves at different times.

- Trees may reach 60 metres or more and form a dense canopy.
- The dense canopy blocks sunlight from reaching the ground.

- They contain rich biodiversity and include trees like rosewood, ebony, mahogany, rubber, and cinchona.
- Animals include elephants, monkeys, leopards, and many birds.
- They are found in the Western Ghats, Northeastern states, and Andaman and Nicobar Islands.

Tropical Deciduous Forests, also called Monsoon Forests, are the most widespread forests in India. They shed leaves for about 6-8 weeks during dry summer to conserve water.

- They are found in regions receiving 70-200 cm rainfall.
- Important trees include sal, teak, peepal, neem, shisham, and sandalwood.
- Animals include tiger, lion, elephant, deer, and birds.
- They occur in Jharkhand, Chhattisgarh, Odisha, Madhya Pradesh, Maharashtra, Karnataka, Himalayan foothills, and parts of the Northeast.
- Moist deciduous forests receive 100-200 cm rainfall, while dry deciduous forests receive 70-100 cm rainfall.

Tropical Thorn Forests and Scrubs are found in areas with low rainfall below 70 cm and high temperature. Vegetation is sparse and adapted to dry conditions.

- Trees have long roots to reach underground water.
- Leaves are small and thick to reduce evaporation.
- Trees include acacia, babul, cactus, khair, and neem.
- Animals include rats, rabbits, foxes, reptiles, and birds.

- They are common in Rajasthan, Gujarat, Haryana, Punjab, and dry parts of Madhya Pradesh, Uttar Pradesh, Karnataka, and Andhra Pradesh.

Montane Forests, also called Himalayan Forests, are found in mountainous areas and change according to altitude. This is called vertical zonation.

- At lower altitudes, tropical and wet temperate forests occur.
- At 1500-3000 m, temperate forests with oak, chestnut, and deodar are found.
- At higher altitudes, coniferous trees such as pine, fir, spruce, silver fir, juniper, and birch occur.
- Animals include snow leopard, Kashmir stag, wild sheep, and mountain birds.
- They are found in the Himalayas, Northeastern hills, and southern hill ranges such as the Nilgiris and Anaimalai.

Mangrove Forests, also called Tidal Forests, grow in coastal areas where rivers meet the sea. They survive in saline water and muddy tidal conditions.

- Mangrove trees have breathing roots or pneumatophores.
- They protect coastlines from erosion, cyclones, and storm surges.
- They provide habitat for fish, crocodiles, turtles, birds, and crabs.
- Important trees include sundari, gewa, mangrove, and palm.
- They are found in the Sundarbans, deltas of Mahanadi, Krishna, and Kaveri, Andaman and Nicobar Islands, and the

Gujarat coast.

3. Wildlife in India

Fauna of India

India has incredible diversity of animal species with nearly 90,000 species of animals. This includes mammals, birds, reptiles, amphibians, fish, insects, and other invertebrates.

- India has many important mammals such as Bengal tiger, Indian elephant, Indian rhinoceros, Asiatic lion, snow leopard, blackbuck, wild ass, and Hangul.
- Bengal Tiger is the national animal of India and is protected in tiger reserves.
- Indian Elephant is the national heritage animal and is found in southern and northeastern states.
- Indian Rhinoceros is found mainly in Kaziranga National Park, Assam.
- Asiatic Lion is found only in Gir Forest, Gujarat.
- Peacock is the national bird of India.

Many species are endangered due to habitat loss, poaching, pollution, and human-wildlife conflict. Protecting wildlife means protecting the ecosystems in which they live.

Flora of India

India has approximately 47,000 species of plants, including flowering plants, trees, herbs, shrubs, grasses, orchids, ferns, and medicinal plants.

- Endemic species are species found only in a particular region. Examples include Nilgiri Tahr, Lion-tailed Macaque, Malabar Civet, and many unique orchids and ferns.
- Medicinal plants are important for systems like Ayurveda, Unani, and Siddha.
- Plants such as neem, tulsi, arjuna, ashwagandha, turmeric, and aloe vera have medicinal value.
- India is known for its rich variety of spice plants and useful forest products.

4. Threats to Forests and Wildlife

Causes of Deforestation

Deforestation is the clearing or destruction of forests. It is one of the greatest threats to biodiversity, climate balance, soil conservation, and tribal livelihoods.

- Agricultural Expansion clears forests for farming and settlement.
- Mining and Quarrying destroy forests in mineral-rich regions such as Jharkhand, Chhattisgarh, and Madhya Pradesh.
- Infrastructure Development such as roads, railways, dams, airports, and housing projects converts forest land.
- Industrialisation requires land and raw materials, increasing pressure on forests.
- Urbanisation expands cities into forest and agricultural land.
- Overgrazing prevents forest regeneration by destroying young saplings.

- Forest Fires destroy vegetation, wildlife habitat, and soil organisms.
- Unsustainable Logging removes trees faster than forests can regenerate.
- Population Pressure increases demand for land, fuelwood, timber, food, and housing.

Threats to Wildlife

- Habitat Loss and Fragmentation is the greatest threat because animals lose shelter, food, breeding grounds, and migration routes.
- Poaching and Illegal Hunting threaten tigers, rhinos, elephants, birds, reptiles, and many rare animals.
- Human-Wildlife Conflict increases when villages, farms, and cities expand into wildlife habitats.
- Pollution affects air, water, soil, rivers, oceans, and food chains.
- Climate Change changes temperature, rainfall, migration, breeding, and habitat suitability.
- Invasive Species such as water hyacinth and lantana disturb native ecosystems.
- Natural Disasters such as floods, droughts, cyclones, and fires destroy habitats.
- Disease can spread from domestic animals to wild populations and reduce species numbers.

5. Conservation of Forests and Wildlife

Conservation Efforts in India

India has taken several steps to conserve forests and wildlife through laws, projects, protected areas, and community participation.

- Project Tiger (1973) was launched to protect the Bengal tiger and its habitat. It began with 9 tiger reserves and has become one of the most important wildlife conservation programmes.
- Project Elephant (1992) protects Indian elephants, their corridors, and their habitats while reducing human-elephant conflict.
- Project Rhino protects the Indian one-horned rhinoceros, especially in Assam.
- National Parks and Wildlife Sanctuaries protect ecosystems where human activity is restricted. Examples include Jim Corbett, Kaziranga, Gir, Periyar, Sundarbans, Ranthambore, and Keoladeo Ghana.
- Biosphere Reserves are large protected areas with core, buffer, and transition zones. Examples include Nilgiri, Nanda Devi, Sundarbans, Gulf of Mannar, Nokrek, and Manas.
- Wildlife Protection Act (1972) provides legal protection to wildlife, prohibits hunting, regulates trade, and establishes protected areas.
- Forest Conservation Act (1980) regulates the diversion of forest land for non-forest use and requires central approval.

- National Forest Policy (1988) emphasises conservation, sustainable use, community participation, and increasing forest cover.
- Compensatory Afforestation Fund Act (2016) provides funds for planting trees when forest land is diverted for development.

Community Participation in Conservation

- Chipko Movement (1973) began in Uttarakhand, where people, especially women, hugged trees to prevent logging. It was associated with leaders like Sundarlal Bahuguna and Chandi Prasad Bhatt.
- Joint Forest Management (JFM) involves local communities in forest protection and benefit-sharing.
- Beej Bachao Andolan in Uttarakhand promoted saving traditional seeds, organic farming, and biodiversity protection.
- Sacred Groves are forest patches protected by communities because of religious beliefs. They are rich in biodiversity and found in states such as Meghalaya, Rajasthan, Maharashtra, and Arunachal Pradesh.
- Appiko Movement in Karnataka was similar to Chipko and protected forests of the Western Ghats from commercial logging.

International Conservation Efforts

- CITES regulates international trade in endangered species.
- Ramsar Convention protects wetlands of international importance.

- World Heritage Convention protects natural sites of outstanding universal value such as Sundarbans, Kaziranga, Nanda Devi, and Western Ghats.
- Convention on Biological Diversity (CBD) aims to conserve biodiversity and ensure sustainable use.
- IUCN maintains the Red List of Threatened Species and works with governments and organisations for conservation.

6. IUCN Red List Categories

Classification of Species Based on Conservation Status

The IUCN Red List classifies species according to their risk of extinction. These categories help governments and conservationists decide which species need urgent protection.

- Normal Species have population levels considered normal for survival, such as common pigeons, rats, cows, and many common plants.
- Endangered Species are in danger of extinction if threats continue. Examples include Bengal tiger, Indian rhinoceros, Asiatic lion, gharial, snow leopard, Indian elephant, and Hangul.
- Vulnerable Species are declining and may become endangered in the future. Examples include blue sheep, sambhar deer, Indian wolf, common langur, and mugger crocodile.
- Rare Species have small populations and may become vulnerable or endangered. Examples include Himalayan

brown bear, wild Asiatic buffalo, dhole, red panda, and lion-tailed macaque.

- Endemic Species are found only in a specific geographic area. Examples include Nicobar pigeon, lion-tailed macaque, Nilgiri tahr, and Malabar civet.
- Extinct Species no longer exist anywhere. Examples include Dodo, passenger pigeon, and Indian cheetah in India.

7. Impact of Biodiversity Loss

Consequences of Biodiversity Loss

Loss of biodiversity has serious consequences for ecosystems, economies, cultures, health, food security, and climate stability.

- Ecological Imbalance occurs when species disappear and food chains are disturbed.
- Loss of Economic Benefits affects agriculture, tourism, forestry, medicine, and industries.
- Reduced Resilience makes ecosystems less able to recover from climate change, disease, and disasters.
- Loss of Cultural Values affects sacred groves, traditional practices, and community identity.
- Food Security Threat arises when pollinators, crop wild relatives, and genetic diversity decline.
- Health Impacts occur when medicinal plants disappear and disease risks increase.
- Soil Erosion and Degradation increase when vegetation cover is removed.

- Climate Change worsens because forests absorb carbon dioxide and store carbon.
- Water Cycle Disruption occurs because forests regulate rainfall absorption, groundwater recharge, and river flow.

8. Sustainable Use of Forest and Wildlife Resources

Principles of Sustainable Use

Sustainable use means using forests and wildlife in a way that meets present needs without harming the ability of future generations to meet their needs.

- Sustainable Logging cuts trees at a rate that allows regeneration.
- Controlled Hunting uses strict rules, permits, quotas, and bans for endangered species.
- Eco-tourism supports conservation and benefits local communities.
- Community-Based Conservation gives local people a role in protecting forests and wildlife.
- Alternative Livelihoods such as beekeeping, handicrafts, eco-tourism, and organic farming reduce dependence on forests.
- Afforestation and Reforestation increase forest cover by planting trees.
- Conservation Breeding increases endangered species populations and may reintroduce them into the wild.

- Habitat Restoration restores degraded ecosystems by removing invasive species and planting native vegetation.

What Can We Do? Individual Actions

- At home, reduce paper, wood, water, fuel, and electricity wastage; recycle paper and avoid products made from endangered species.
- In daily life, plant trees, join awareness campaigns, support conservation organisations, and report illegal wildlife trade.
- When travelling, follow protected-area rules, do not disturb wildlife, avoid wildlife products, and support eco-friendly tourism.
- As a community, organise cleanliness drives, support local conservation efforts, and participate in forest protection programmes.

Key Statistics About Forests in India

- Total forest and tree cover is about one-fourth of India's geographical area.
- Forest cover alone is about one-fifth of the geographical area.
- National Forest Policy target is 33% forest cover.
- Madhya Pradesh has the largest forest area.
- Arunachal Pradesh has one of the highest percentages of forest cover.
- Tiger population has increased significantly due to conservation programmes and tiger reserves.